Sustainable Water Future: Predicting Potability Using Machine Learning

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Introduction

Access to clean and drinkable water is essential for human health and environmental sustainability.
In this project, we used data mining and classification techniques to predict water potability based on its physical and chemical properties such as pH, Hardness, Solids, and Chloramines.
The original dataset was imbalanced, containing more non-potable samples than potable ones.
To address this, we generated synthetic data using AI-based balancing techniques to create a balanced dataset, which improved model learning and accuracy.
This research supports environmental sustainability by applying artificial intelligence and machine learning techniques to enhance water quality prediction and ensure access to clean and safe water for future generations.

VISION 2030
WATER SUSTAINABILITY

Methodology

1. Business Understanding: The study analyzes factors affecting water potability using the CRISP-DM data-mining framework.
2. Data Preparation: Data was processed in RapidMiner by removing duplicates, handling missing values, and balancing records.
3. Feature Selection: The Select Attributes operator was applied to keep the most relevant features.
4. Modeling: Three models Decision Tree, Random Forest, and GBOOST were built in RapidMiner to predict water potability.
5. Evaluation: Models were tested using cross-validation and compared by accuracy, precision, and recall.
6. Deployment: The best model was identified for possible future use in water-quality prediction.

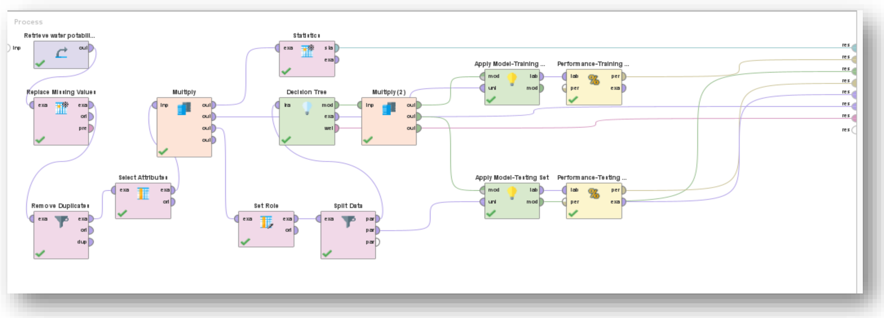

Figure1. Process Decision Tree model in RapidMiner.

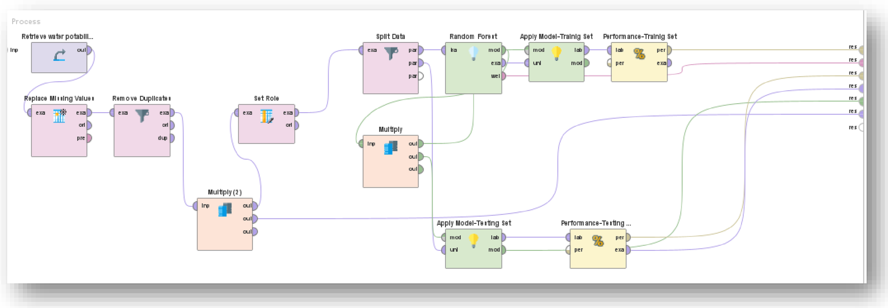

Figure 2. Process of creating the Random Forest model.

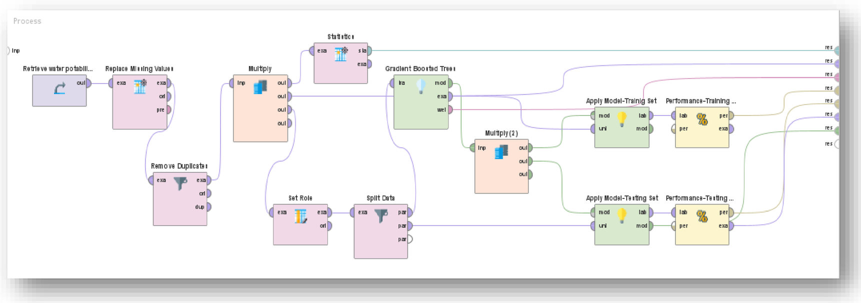

Figure 3. Process of training the Gradient Boosted Trees model

Scientific Contribution and Objectives

This project focuses on predicting water potability using data-mining and machine-learning techniques to support environmental sustainability and efficient water resource management.
The main goal is to classify water samples as potable or non-potable based on several physical and chemical attributes such as pH, hardness, solids, and chloramines.
Three classification models — Decision Tree, Random Forest, and Gradient Boosted Trees (GBOOST) — were applied to compare their accuracy and stability.

Beyond technical modeling, this study highlights the importance of leveraging AI-based data analytics to support Saudi Arabia's national sustainability goals.
The results emphasize how data balancing and feature selection can significantly enhance model performance, demonstrating the potential of AI in water-quality prediction.
This research aligns with Vision 2030 and the RDIA's objectives by providing insights that promote data-driven decisions for sustainable water management.

Water consumption across sectors in Saudi Arabia shows that agriculture uses the largest share of total water resources. This emphasizes the importance of improving water quality and promoting sustainable water management to support Vision 2030.

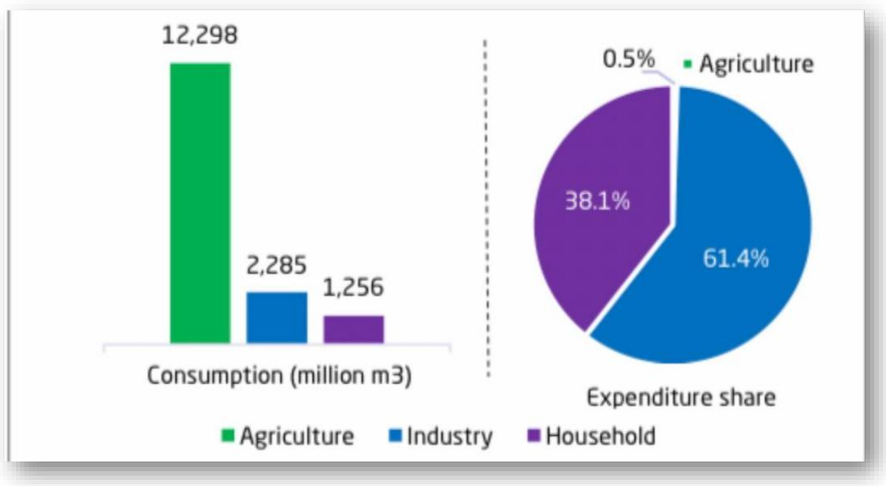

Figure 1. Water consumption and expenditure share by sector in Saudi Arabia (2023). Source: GASTAT.

This figure shows the distribution of water sources in Saudi Arabia, highlighting the increasing reliance on desalinated water and the decline in non-renewable groundwater. This emphasizes the need for sustainable water management and supports the project's focus on environmental sustainability.

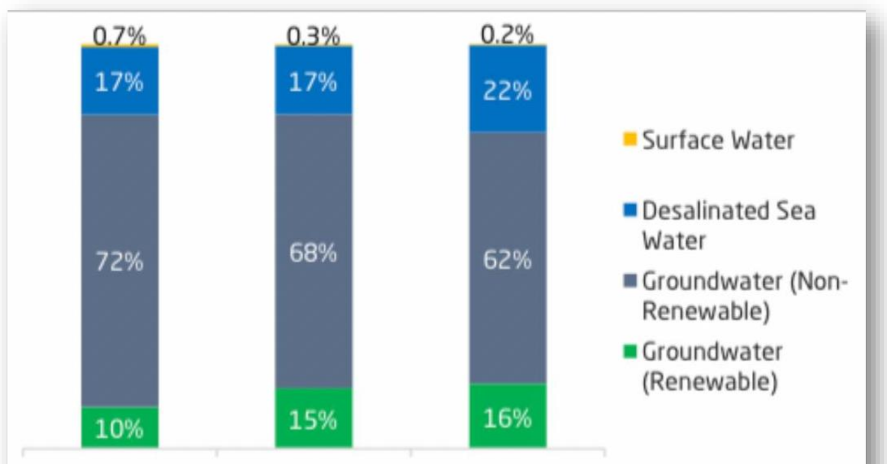

Figure 2. Water supply by source from natural resources in Saudi Arabia (2021 – 2023). Source: GASTAT.

Results

Attribut...	ph	Hardness	Solids	Chlora...	Sulfate	Conduc...	Organic...	Trihalo...	Turbidity	Potability
ph	1	0.056	-0.052	0.004	-0.020	0.007	0.027	0.000	-0.024	0.004
Hardness	0.056	1	-0.034	-0.030	-0.076	-0.023	0.009	-0.002	-0.008	-0.013
Solids	-0.052	-0.034	1	-0.086	-0.204	0.012	-0.003	-0.003	0.019	0.041
Chlorami...	0.004	-0.030	-0.086	1	0.020	-0.019	-0.016	0.023	-0.012	0.024
Sulfate	-0.020	-0.076	-0.204	0.020	1	-0.027	0.039	-0.046	-0.014	-0.039
Conducti...	0.007	-0.023	0.012	-0.019	-0.027	1	0.033	0.002	-0.004	0.009
Organic...	0.027	0.009	-0.003	-0.016	0.039	0.033	1	-0.014	-0.038	-0.039
Trihalom...	0.000	-0.002	-0.003	0.023	-0.046	0.002	-0.014	1	-0.026	0.014
Turbidity	-0.024	-0.008	0.019	-0.012	-0.014	-0.004	-0.038	-0.026	1	0.004
Potability	0.004	-0.013	0.041	0.024	-0.039	0.009	-0.039	0.014	0.004	1

The correlation matrix illustrates the strength of relationships among the water quality attributes. Attributes such as pH, Solids, and Chloramines had stronger correlations with Potability, indicating their higher influence in predicting drinkable water.

The correlation matrix shows the relationships between water quality attributes and potability, highlighting pH, Solids, and Chloramines as the most influential factors.

We first used the Decision Tree, which showed clear overfitting between training and testing results.
To reduce this, the Select Attributes operator was applied, which slightly improved the model's performance but the overfitting issue remained.
The Random Forest provided more stable results and reduced the gap between training and testing.
Finally, GBOOST achieved the best accuracy of 72.05%, with balanced and consistent results, and was selected as the final model for predicting water potability.

accuracy: 53.40% +/- 1.90% (micro average: 53.40%)

	true 0	true 1	class precision
pred. 0	1329	1234	51.85%
pred. 1	70	165	70.21%
class recall	95.00%	11.79%	

Figure 2. Model performance results (Random Forest)

accuracy: 50.11% +/- 0.42% (micro average: 50.11%)

	true 0	true 1	class precision
pred. 0	1390	1387	50.05%
pred. 1	9	12	57.14%
class recall	99.36%	0.86%	

Figure 1. Model performance results (Decision Tree)

accuracy: 72.05% +/- 2.25% (micro average: 72.05%)

	true 0	true 1	class precision
pred. 0	837	220	79.19%
pred. 1	562	1179	67.72%
class recall	59.83%	84.27%	

Figure 3. Model performance results (GBOOST)

Conclusion and Policy Implications

This study demonstrates that machine learning and data mining techniques can effectively predict water potability when supported by proper data cleaning, AI-based balancing, and model optimization.
By generating balanced data using artificial intelligence, the accuracy and fairness of the models significantly improved.
Among all tested algorithms, the Gradient Boosted Trees (GBOOST) model achieved the best performance, offering stable and consistent results for water quality classification.